The Socio-Economic Consequences of Housing Assistance

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CF/HH Reading Group
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Motivation

The Paper in a nutshell:

This paper analyzes the effect of Europe's largest public housing program on houshold's socio-economic outcomes.

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- **2015-2025**: House prices in the EU increased by 64% and rents by 21% according (Eurostat).
- **European Affordable Housing Plan**
- The initiative aims to address one of the most urgent concerns for citizens: access to affordable, sustainable and quality housing.

Literature

Experimental Evidence & Programs:

- Jacob et al. (2015); Barnhardt et al. (2016); Jacob and Ludwig (2012)
- Kling et al (2004); Ludwig et al. (2013); Mills et al. (2006); Gubits et al. (2016)

In-kind Transfers & Labor Supply:

- Murray (1980); Leonesio (1988); Gahvari (1994); Shroder (2002)
- Moffitt (2002); Currie and Gahvari (2008)

Lotteries & Cutoffs Embedded:

- Angrist (1990), Angrist and Krueger (1991); Angrist et al. (1996); Imbens et al. (2001); Kling et al. (2007)

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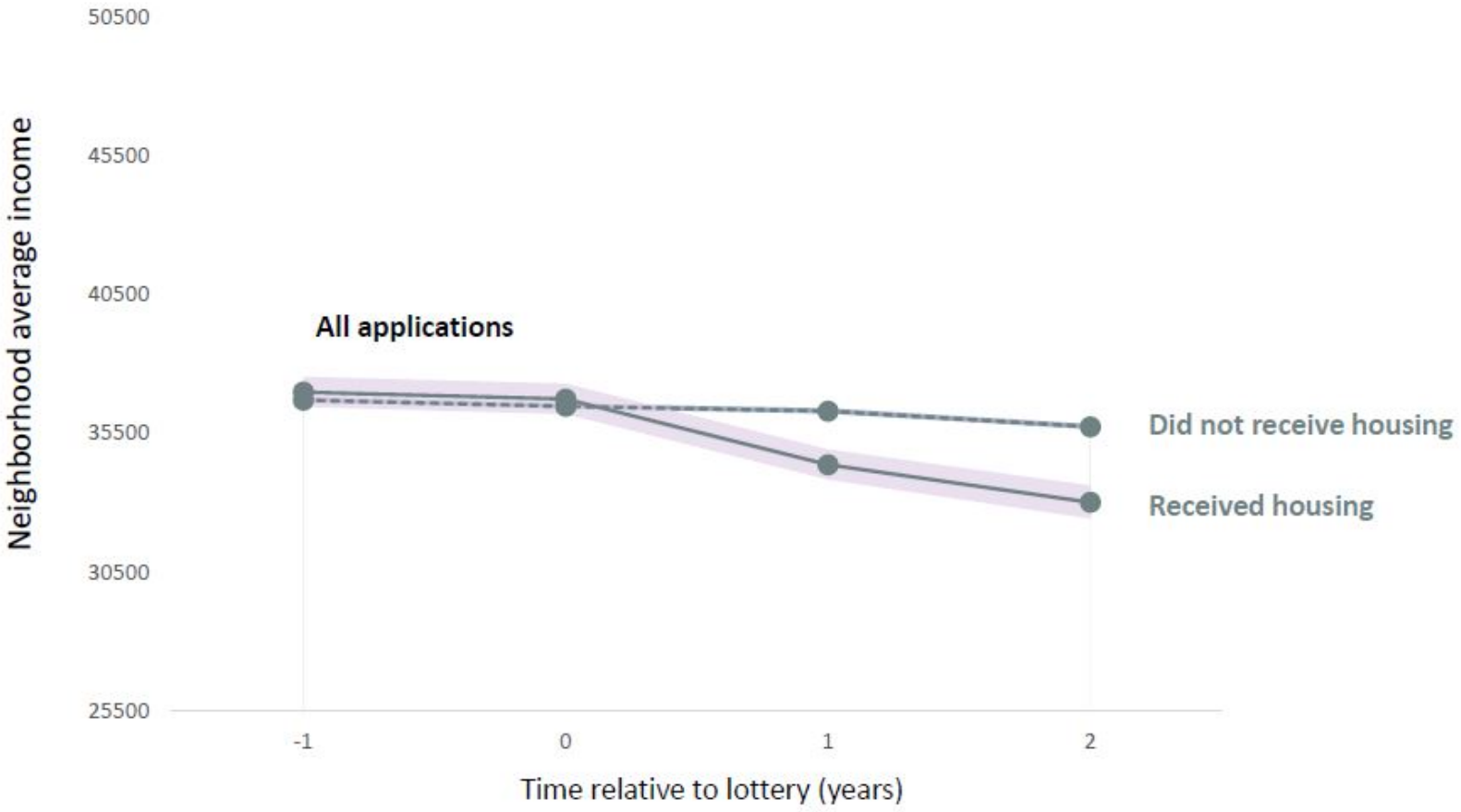
Average vs. Heterogeneous Effects:

- Public housing is associated with **negative** labor and neighborhood outcomes
- **Heterogeneity**: effects depend on neighborhood income and quality: *upward* or *dowards* moves.

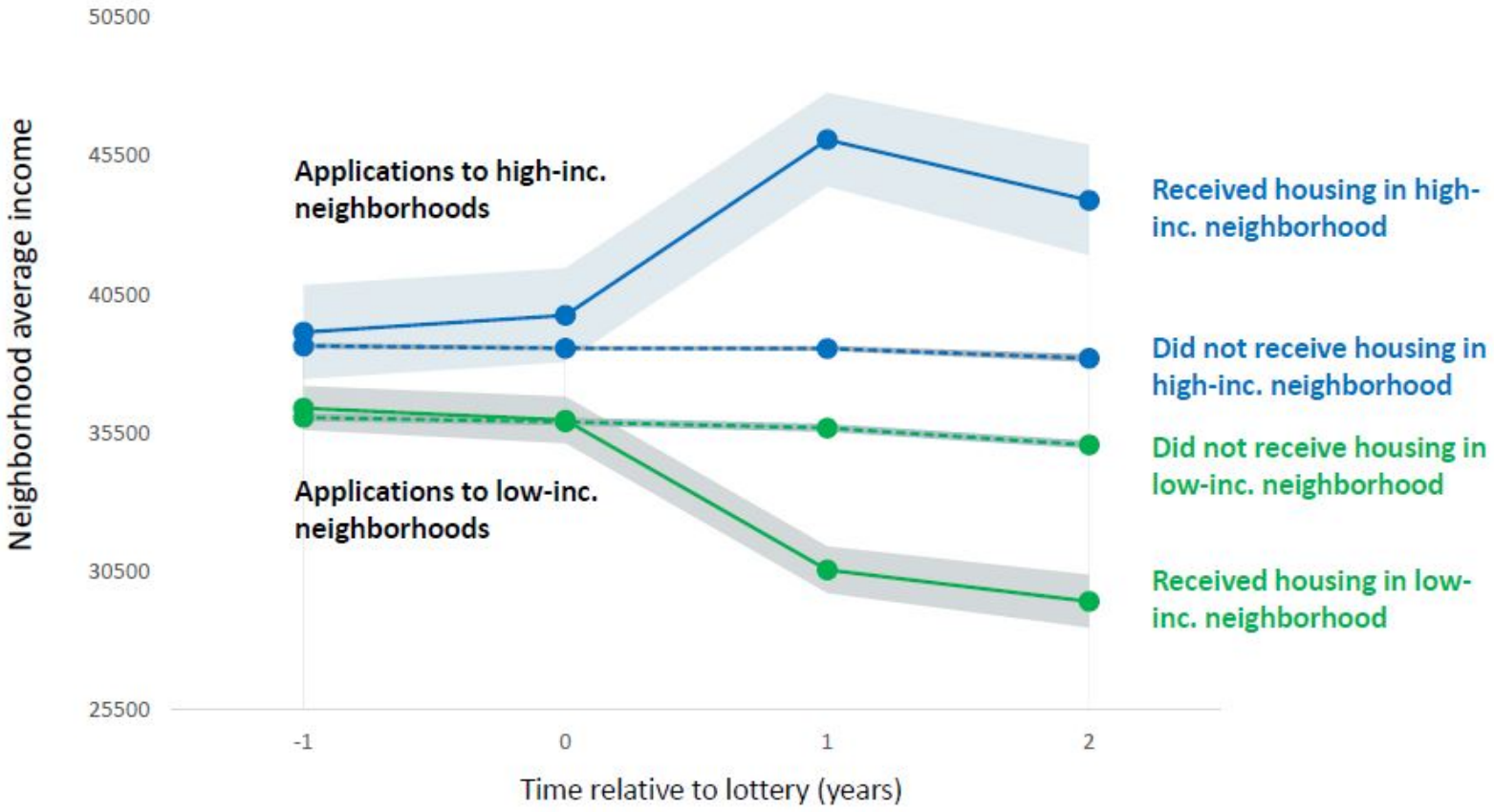
Heterogeneity in lottery-induced moves

(a) On average, across lotteries.



Heterogeneity in lottery-induced moves

(b) Lotteries in high-income vs. low-income neighborhoods.



Institutional Setting

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Public Housing in Amsterdam Metropolitan Area:

- 46% of housing stock
- Eligibility: household size-adjusted median income.
- Type of Housing: high-rise apartment buildings (located far from city) and apartments in high-rent urban areas

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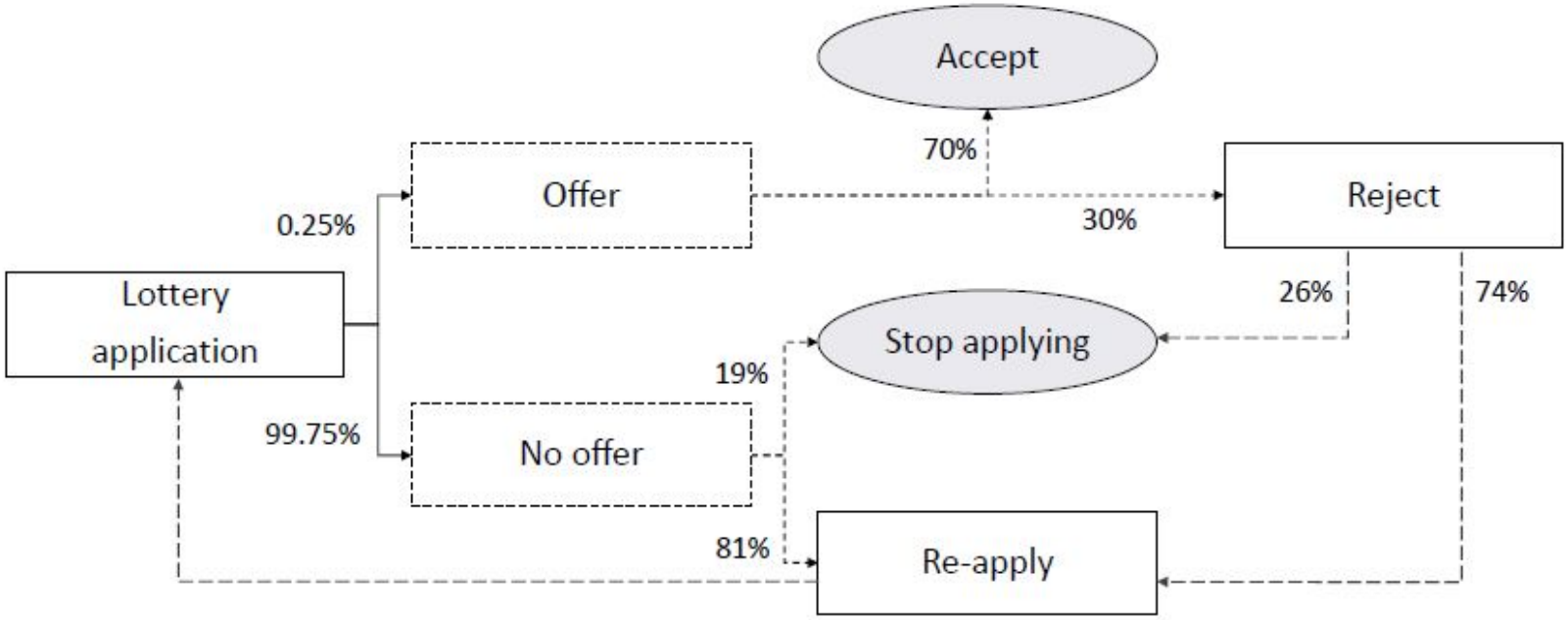
Allocation System:

- Waiting list (very long waits) $\approx$ 12 years!
- Housing lotteries for some small sample of units

Online Platform Amsterdam: **WoningNet**
Online Platform Tilburg and Breda: **WonenBreburg**

The lottery process



- Each lottery: very low probability of winning.
- The probability of future wins is very low.
- Maximum of 2 applications max per week.
- No receiving an offer or reject an offer does not prevent participation in future application periods.

Data

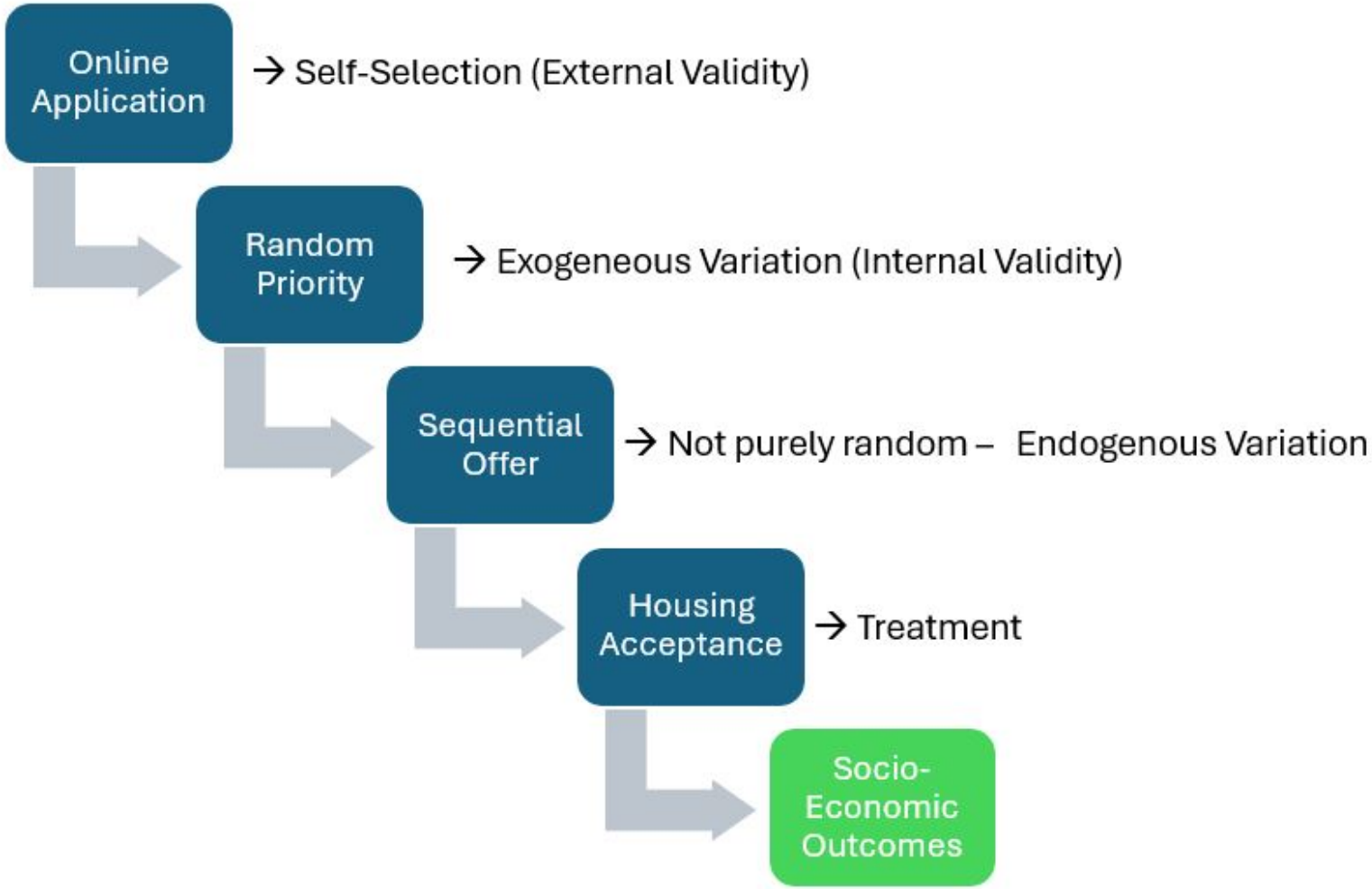
Data sources:

- Amsterdam housing associations: *Platform Woningcorporaties Noordvleugel Randstad* (online platform WoningNet)
- Statistics Netherlands Microdata (CBS)
  - Address histories from *Municipal registers*
  - Individuals' annual earnings from *Tax Returns*
  - Public assistance receipt from *Agencies*

Data merging:

- **Individual**: date of birth, gender, and address
  $\Rrightarrow$ 96% match rate
- **Housing units**: address
  $\Rrightarrow$ 99% match rate

Selection, Randomization, and Endogeneity in the Lottery Process



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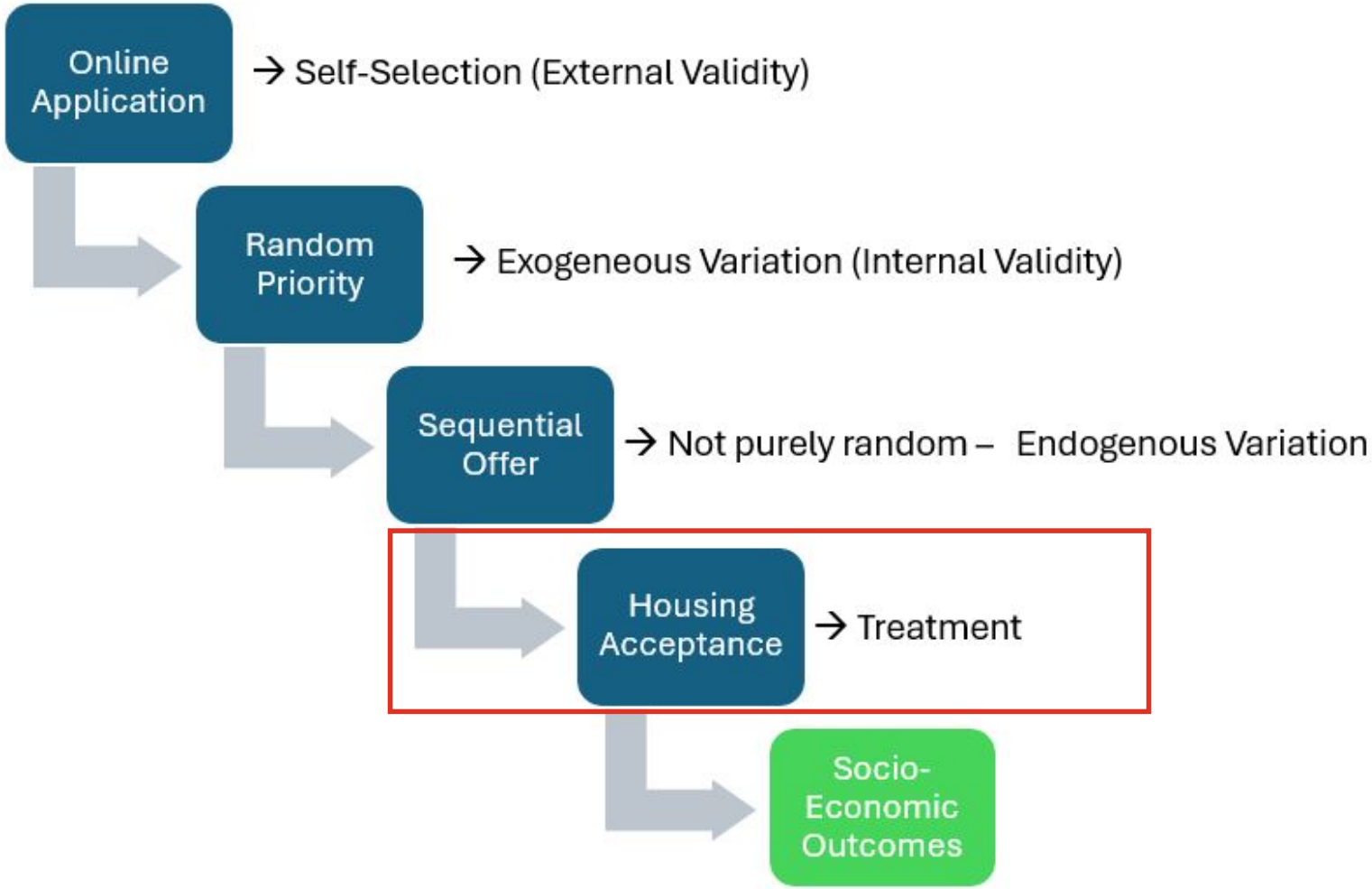

Treatment & Counterfactual

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- Receiving public housing VIA the lottery during study period (compliers)

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- Losing applicants can re-apply in future lotteries
- Moves them partially toward the treatment, creating small bias
  $\Rrightarrow$ Probability of receiving an offer in any lottery is very low.

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Substitution bias:

- Some non-winners may get housing via the **waiting list**
  $\Rrightarrow$ Wait list is highly over-subscribed
- Substitution rates are similar across neighborhood

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$\Rrightarrow$ **Nearly Zero Bias Claim**: Cumulative bias over multiple lotteries? If policymakers scale the program significantly? Especially in neighborhoods with high turnover.

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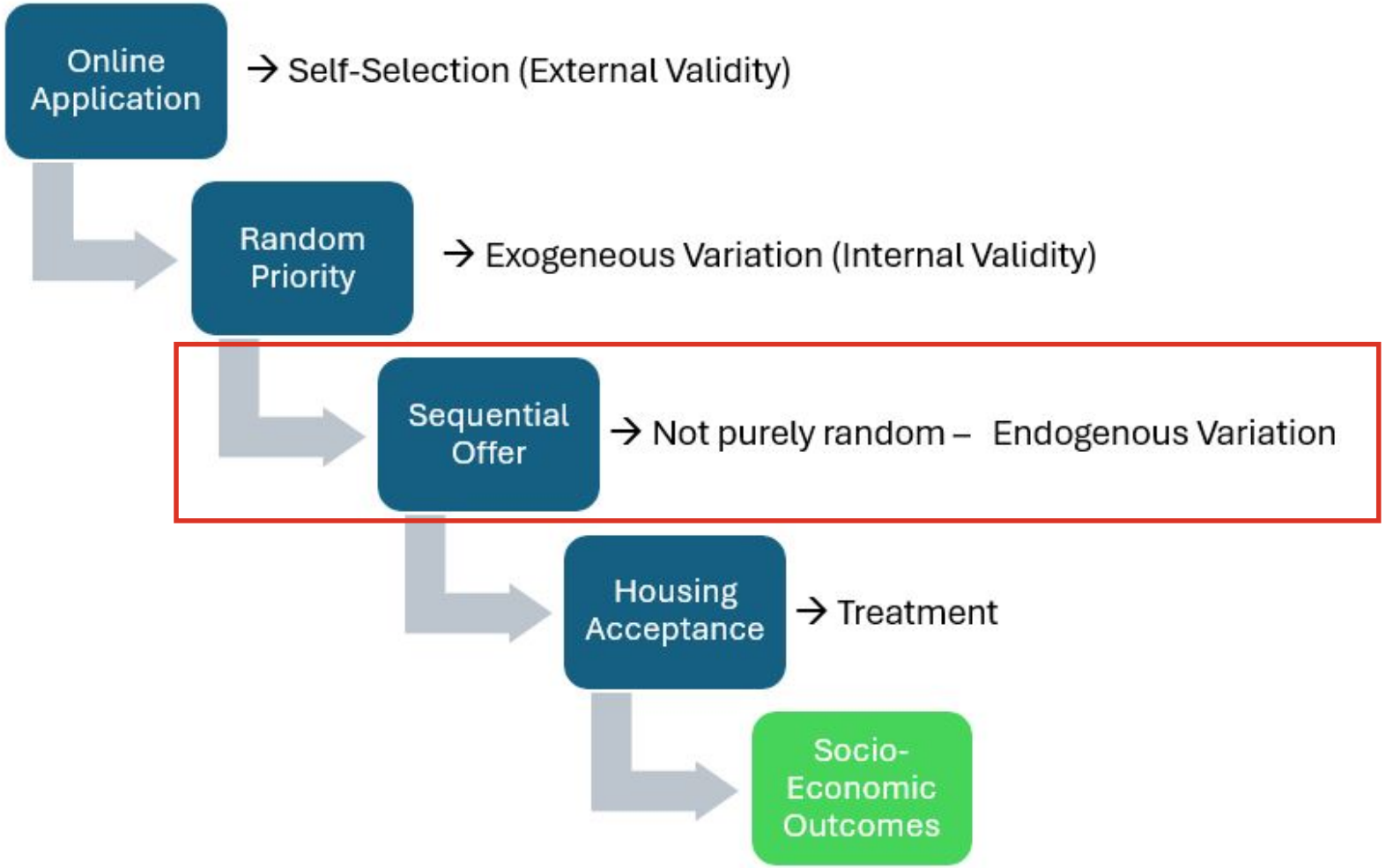

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Case 1

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Offer assignment $\neq$ Not purely random

Empirical Design

Consider lotteries indexed by j, and applicants to lottery j indexed by i.

For a given lottery j:

- N_j: number of applicants
- Z_{ij}: indicator for applicant i receiving an offer
- $D_{ij}(z)$, $z \in \{0, 1\}$: treatment status given offer
- $Y_{ij}(d)$, $d \in \{0, 1\}$: potential outcomes

Two-stage least squares specification:

$$D_{ij} = \gamma + \delta Z_{ij} + L_j + \nu_{ij} \tag{1}$$

$$Y_{ij} = \alpha + \beta D_{ij} + L_j + \varepsilon_{ij} \tag{2}$$

where L_j is a lottery fixed effect.

Key assumption: random assignment of offers.

$$(D_{ij}(0), D_{ij}(1), Y_{ij}(0), Y_{ij}(1))_{i \in \{1,\ldots,N_j\}} \perp\!\!\!\perp Z_j \quad \forall j$$

$\Rrightarrow$ But, Z_{ij} is a function of $D_{ij}(1)$!

Offer Assignment and Priority Structure

Violation of *assumption random assignment of offers*

- Random assignment of offers is *not* satisfied in this setting, since Z_j depends on $D_j(1)$, the random vector of complier statuses.

Assumption random assignment of priority

- Let P_j collect priorities $(P_{ij})_{i=1}^{N_j}$ in a vector.
- Then $Z_{ij} = 1$ for all applicants assigned a priority weakly higher than the highest-priority complier, and zero otherwise:

$$Z_{ij} = \mathbf{1}\left\{ P_{ij} \leq \min_{\substack{k \in \{1,\ldots,N_j\} \\ D_{kj}(1)=1}} P_{kj} \right\}, \quad i \in \{1, \ldots, N_j\}.$$

(A1) Random assignment of priority, by construction, Pj is a randomly drawn

(A2) Relevance (60% of the first offers are accepted)

(A3) One-sided non-compliance

Toy Example: Housing Lottery

Setup: One lottery with $N_j = 5$ applicants

Applicant	Priority P_{ij}	Offer Z_{ij}	Accept D_{ij}	Earnings Y_{ij}
C	1	1	0	20
A	2	1	0	22
E	3	1	1	18
B	4	0	0	25
D	5	0	0	24

Allocation rule:

- Random priority order P_{ij} is assigned
- Offers are made sequentially
- Process stops at the first applicant who accepts

- First accepter: Applicant E ($P_{ij} = 3$)
- Applicants with higher priority receive offers

$$Z_{ij} = \mathbf{1}\{P_{ij} \leq 3\}$$

Resulting variation:

$$\mathbf{Z}_j = (1, 1, 1, 0, 0), \quad \mathbf{D}_j = (0, 0, 1, 0, 0)$$

Results

Negative effects on:

- Employment
- Individual and household earnings
- Neighborhood income and house values

Positive effects on:

- Public assistance receipt

Interpretation:

- Recipients tend to work less, earn less, and rely more on government support

Key limitation:

- Average effects **obscures substantial heterogeneity** across lotteries

Effects of public housing receipt: two-year impacts.

	Base	OLS (1)	IV (Z_{ij}) (2)	N
I. *Income and employment*				
HH labor income	17,711	-2,279***	-2,375***	439,381
		(647.3)	(871.8)	
HH total income	22,834	-1,371**	-1,323*	439,381
		(547.2)	(751.7)	
HH disposable income	20,129	-1,028**	-1,042*	439,381
		(440.6)	(615.0)	
Individual labor income	17,042	-1,795***	-1,513*	467,027
		(635.7)	(615.0)	
Individual total income	22,040	-810.1	-382.9	467,027
		(542.2)	(729.1)	
Employment	0.58	-0.0440***	-0.0341	439,877
		(0.0169)	(0.0227)	
II. *Housing quality and rent expenditure*				
House value (per m^2)	2,635	-105.5***	-154.5***	463,373
		(30.50)	(43.32)	
Monthly rent expenditure (net of subsidy)	445.8	10.70	-5.998	464,181
		(13.48)	(19.11)	
Cum. two-year residential move rate	0.51	-0.2754***	-0.2987***	467,027
		(0.0168)	(0.0239)	
III. *Neighborhood characteristics*				
NH average income	36,470	-2,913***	-3,359***	467,027
		(237.3)	(361.7)	
NH fraction non-western	0.24	0.0505***	0.0533***	467,027
		(0.004)	(0.0084)	
Social distance	0.50	0.0404***	0.0349***	467,027
		(0.0067)	(0.0085)	
Distance to city center (m)	11,752	-3,555***	-3,975***	467,027
		(483.2)	(683.2)	
IV. *Fiscal cost*				
Public assistance receipt	0.23	0.0540***	0.0367**	439,877
		(0.0140)	(0.182)	
Fiscal externality	-2,456	-1,207***	-1,160***	467,027
		(294.9)	(401.0)	
Lottery FE		✓	✓	

Results: Heterogeneous Effects

Across neighborhoods:

- Moving to high-income neighborhoods:
  $\Rrightarrow$ Increases individual and household income
- Moving to low-income neighborhoods:
  $\Rrightarrow$ Decreases income outcomes

Within neighborhoods:

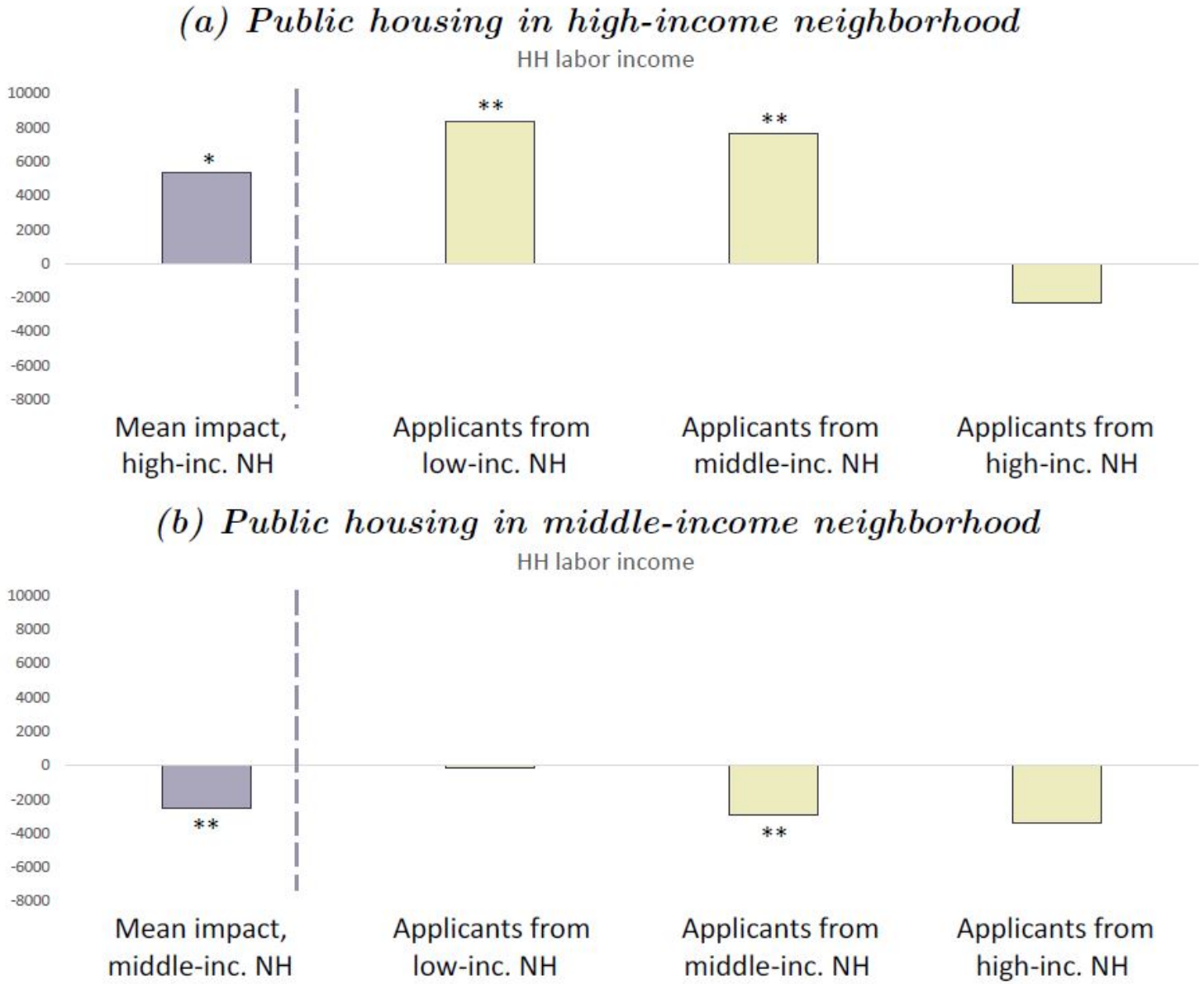
- Positive effects are driven by *upward moves*
- Negative effects are driven by *downward moves*

Main takeaway:

- The impact of public housing depends critically on **neighborhood quality and direction of the move**

Treatment on the treated, by neighborhood income of public housing unit

...by neighborhood income of applicant at time of application



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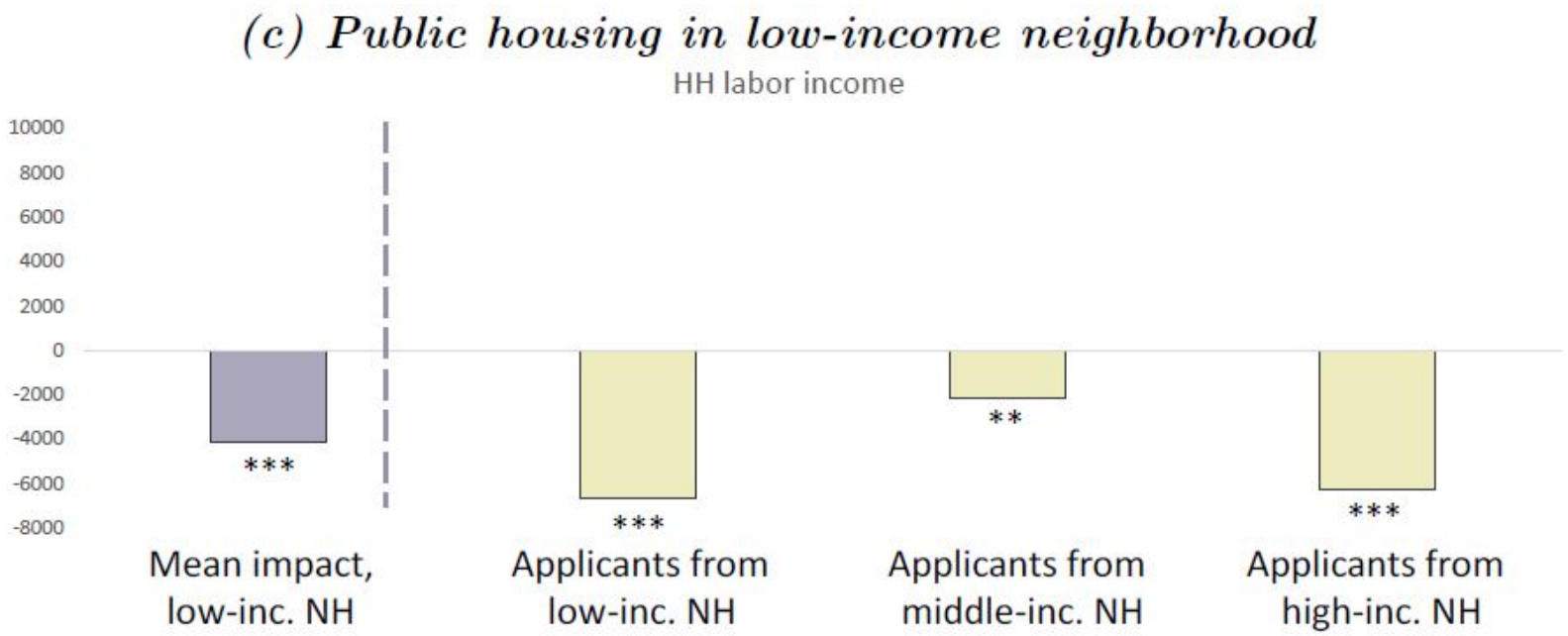

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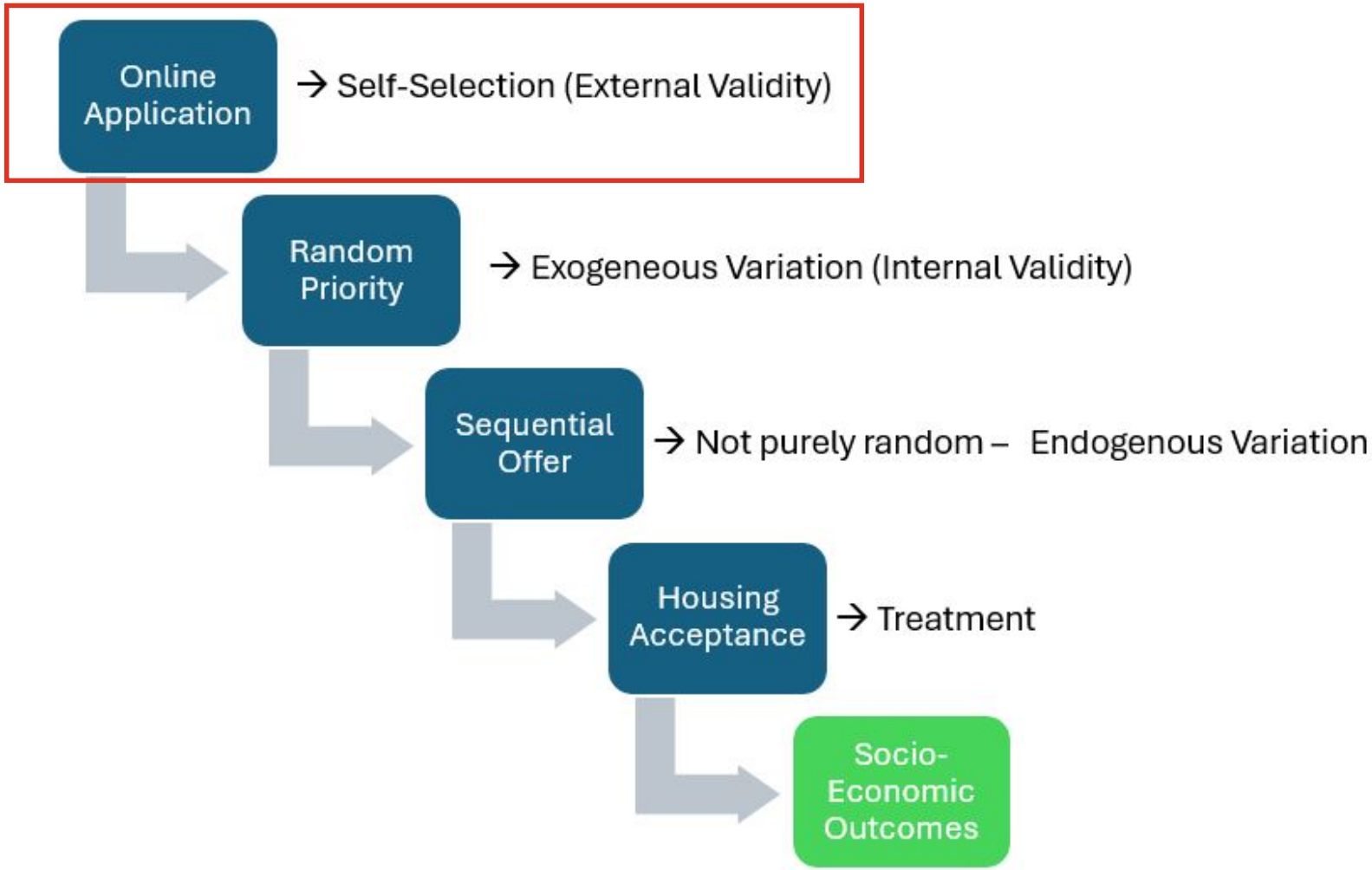

Self-selection into lotteries

Selection into lotteries:

- **Repeatedly self-select into lotteries:** Potential applicants in the Amsterdam housing lotteries will not in general prefer all of the public housing units.
- **Applicants pools:** Applicants with lower lower baseline neighborhood income tend to apply to low-income neighborhoods.
  $\Rrightarrow$ Stronger preferences for leisure and non-housing consumption
- **Strategic incentives:** Housing in low-income neighborhoods receive substantially fewer applications.
  $\Rrightarrow$ The lottery creates incentives for more immediate need of housing.

$\Rrightarrow$ Affects **external validity**

Dynamic discrete choice mode

- The model explains application behavior
- Little evidence that applicants select based on unobserved gains

$\Rrightarrow$ Policies placing low-income households in high-income neighborhoods could improve outcomes.

Conclusion

- **Public Housing Reduces Labor Supply on Average:**
- **Neighborhood Effects are Crucial:**
  - Placement in high-income neighborhoods can improve earnings and employment outcomes
  - Conversely, remaining in low-income neighborhoods can reinforce negative labor market outcomes
  - Highlights the importance of spatial targeting in housing policies
- **Policy Recommendations:**
  - Design housing programs that consider both household needs and neighborhood quality
- **Broader Implications:**
  - Housing policy is not just about affordability, it affects labor market outcomes and social mobility